

HomeWork-2

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Problem 1)

1.1) We plug the model $Y_i = X_i \beta + \epsilon_i$ into the OLS formula & simplify. Then we take conditional expectation given $X_1, \dots, X_n$. Since the errors have mean 0 & are independent of the X_i 's, the extra term disappears.

$$\hat{\beta} = \beta + \left(\sum_{i=1}^n X_i^2 \right)^{-1} \sum_{i=1}^n X_i \epsilon_i$$

and $E[\hat{\beta} | X_1, \dots, X_n] = \beta$.

→ For eg, OLS is true β plus a noise term. Since that has mean 0, the estimator is unbiased.

$$\hat{\beta} = \left(\sum_{i=1}^n X_i^2 \right)^{-1} \sum_{i=1}^n X_i Y_i$$

Substitute $Y_i = X_i \beta + \epsilon_i$

$$\hat{\beta} = \left(\sum_{i=1}^n X_i^2 \right)^{-1} \sum_{i=1}^n X_i (X_i \beta + \epsilon_i)$$

$$= \left(\sum_{i=1}^n x_i^2 \right)^{-1} \left(\sum_{i=1}^n x_i^2 \beta + \sum_{i=1}^n x_i \epsilon_i \right)$$

$$= \beta + \left(\sum_{i=1}^n x_i^2 \right)^{-1} \sum_{i=1}^n x_i \epsilon_i$$

→ Now take conditional expectation

$$E[\hat{\beta} | x_1, \dots, x_n] =$$

$$E\left[\beta + \left(\sum_{i=1}^n x_i^2\right)^{-1} \sum_{i=1}^n x_i \epsilon_i \mid x_1, \dots, x_n\right]$$

$$= \beta + \left(\sum_{i=1}^n x_i^2\right)^{-1} \sum_{i=1}^n x_i E[\epsilon_i | x_1, \dots, x_n]$$

$$\text{Since } E[\epsilon_i | x_1, \dots, x_n] = 0$$

$$E[\hat{\beta} | x_1, \dots, x_n] = \beta$$

$$\therefore \hat{\beta} = \beta + \left(\sum_{i=1}^n x_i^2\right)^{-1} \sum_{i=1}^n x_i \epsilon_i$$

$$\therefore E[\hat{\beta} | x_1, \dots, x_n] = \beta$$

1.2) For ridge, the denominator becomes $\sum x_i^2 + \lambda$. After plugging the model, the coefficient of β is no longer 1. That is why ridge is biased when $\lambda > 0$.

$$\hat{\beta}_\lambda = \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2 + \lambda} \beta + \frac{\sum_{i=1}^n x_i \epsilon_i}{\sum_{i=1}^n x_i^2 + \lambda}$$

and

$$E[\hat{\beta}_\lambda | x_1, \dots, x_n] = \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2 + \lambda} \beta \neq \beta$$

→ For eg, Ridge shrinks the estimate toward 0, so its avg is not exactly β .

→ For ridge

$$\hat{\beta}_\lambda = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2 + \lambda}$$

Substitute $Y_i = X_i \beta + \epsilon_i$

$$\hat{\beta}_\lambda = \frac{\sum_{i=1}^n X_i (X_i \beta + \epsilon_i)}{\sum_{i=1}^n X_i^2 + \lambda}$$

$$= \frac{\sum_{i=1}^n X_i^2}{\sum_{i=1}^n X_i^2 + \lambda} \beta + \frac{\sum_{i=1}^n X_i \epsilon_i}{\sum_{i=1}^n X_i^2 + \lambda}$$

Now Taking conditional Expectation

$$E[\hat{\beta}_\lambda | X_1, \dots, X_n] = \frac{\sum_{i=1}^n X_i^2}{\sum_{i=1}^n X_i^2 + \lambda} \beta + \frac{1}{\sum_{i=1}^n X_i^2 + \lambda} E$$

$$\left[\sum_{i=1}^n X_i \epsilon_i | X_1, \dots, X_n \right]$$

Since $E[\epsilon_i | X_1, \dots, X_n] = 0$

$$E[\hat{\beta}_\lambda | X_1, \dots, X_n] = \frac{\sum_{i=1}^n X_i^2}{\sum_{i=1}^n X_i^2 + \lambda} \beta$$

for $\lambda > 0$

$$\frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2 + \lambda} \neq 1$$

$$\& E[\hat{\beta}_\lambda | x_1, \dots, x_n] \neq \beta$$

This makes sense because ridge adds shrinkage, so it pulls the estimate away from the true value on average.

1.3) Both OLS & ridge can be written as constants plus weighted sums of the errors ϵ_i . The problem gives the fact that a weighted sum of Gaussian errors is again Gaussian with variance $\|a\|_2^2 \sigma^2$. Using that fact gives the variance formula directly.

$$\text{Var}(\hat{\beta} | x_1, \dots, x_n) = \frac{\sigma^2}{\sum_{i=1}^n x_i^2}$$

and

$$\text{Var}(\hat{\beta}_\lambda | x_1, \dots, x_n) = \frac{1}{\left(1 + \frac{\lambda}{\sum_{i=1}^n x_i^2}\right)^2} \cdot \frac{\sigma^2}{\sum_{i=1}^n x_i^2}$$

As λ increases the ridge variance decreases.

→ When $\lambda = 0$, ridge is the same as OLS.
As λ gets bigger, the variance becomes smaller.

$$\hat{\beta} = \beta + \left(\sum_{i=1}^n x_i^2 \right)^{-1} \sum_{i=1}^n x_i \varepsilon_i$$

So

$$\text{Var}(\hat{\beta} | x_1, \dots, x_n) = \text{Var} \left(\left(\sum_{i=1}^n x_i^2 \right)^{-1} \sum_{i=1}^n x_i \varepsilon_i \mid x_1, \dots, x_n \right)$$

$$= \left(\sum_{i=1}^n x_i^2 \right)^{-2} \text{Var} \left(\sum_{i=1}^n x_i \varepsilon_i \mid x_1, \dots, x_n \right)$$

$$\text{Var} \left(\sum_{i=1}^n x_i \varepsilon_i \mid x_1, \dots, x_n \right) = \sigma^2 \sum_{i=1}^n x_i^2$$

Therefore,

$$\text{Var}(\hat{\beta} | x_1, \dots, x_n) = \left(\sum_{i=1}^n x_i^2 \right)^{-2} \left(\sigma^2 \sum_{i=1}^n x_i^2 \right) = \frac{\sigma^2}{\sum_{i=1}^n x_i^2}$$

$$\rightarrow \hat{\beta}_{\lambda} = \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2 + \lambda} \beta + \frac{\sum_{i=1}^n x_i \varepsilon_i}{\sum_{i=1}^n x_i^2 + \lambda}$$

$$\text{Var}(\hat{\beta}_\lambda | x_1, \dots, x_n) = \text{Var}\left(\frac{\sum_{i=1}^n x_i \varepsilon_i}{\sum_{i=1}^n x_i^2 + \lambda} \mid x_1, \dots, x_n\right)$$

$$= \frac{1}{\left(\sum_{i=1}^n x_i^2 + \lambda\right)^2} \text{Var}\left(\sum_{i=1}^n x_i \varepsilon_i \mid x_1, \dots, x_n\right)$$

$$= \frac{1}{\left(\sum_{i=1}^n x_i^2 + \lambda\right)^2} \left(\sigma^2 \sum_{i=1}^n x_i^2\right)$$

$$= \frac{\sigma^2 \sum_{i=1}^n x_i^2}{\left(\sum_{i=1}^n x_i^2 + \lambda\right)^2}$$

$$= \frac{1}{\left(1 + \frac{\lambda}{\sum_{i=1}^n x_i^2}\right)^2} \cdot \frac{\sigma^2}{\sum_{i=1}^n x_i^2}$$

Since the denominator increases as λ increases, the variance decreases.

This makes sense because ridge shrinkage makes the estimator less spread out.

Problem-2)

2.1) We use the condition the problem gives for the minimizer. Then we expand the expectation & solve for $\hat{m}$. That directly gives the answer.

$$\arg \min_m E[(Y - m)^2] = E[Y]$$

→ If the avg value of Y is 5, then the constant predictor with the smallest mean squared error is 5.

→ We are given minimizer $\hat{m}$

$$E[2(Y - \hat{m})] = 0.$$

$$2E[Y - \hat{m}] = 0 \quad \text{Linearity Expectation}$$

$$2(E[Y] - \hat{m}) = 0$$

$$E[Y] - \hat{m} = 0$$

$$\hat{m} = E[Y]$$

$$\therefore \arg \min_m E[(Y - m)^2] = E[Y]$$

This makes sense because among all constant predictions the mean is the one that minimizes squared root.

2.2) Now instead of choosing one constant number, we choose a function of X . Once X is fixed, $f(x)$ acts like a constant for the conditional problem. Then part (d) says the best such value is the conditional mean of Y given X . Finally we use total expectation.

$$\arg \min_{f(x)} E[(Y - f(x))^2]$$

$$\arg \min_{f(x)} E[(Y - f(x))^2] = E[Y | x]$$

If for people with $X = x$, the avg value for Y is 10, then the best prediction for that group is 10.

$$E[(Y - f(x))^2]$$

$$E[(Y - f(x))^2] = E[E[(Y - f(x))^2 | x]]$$

Now fix x , then $f(x)$ is just a constant inside the conditional expectation.

$$E[(Y - f(x))^2 | x]$$

$$f(x) = E[Y | x]$$

$$\arg \min_{f(x)} E[(Y - f(x))^2] = E[Y | x]$$

Problem 3)

3.3) Cross-validation is needed because it tells us which method is likely to predict better on new unseen movies, not just on the data used to fit the model. If we only compare training-error, the more flexible method can look better simply because it fits the training data more closely, even if it doesn't generalize well. We know training error can be biased toward overfit models, & that validation data should be used to select the best model before testing it on the test set. In our results, a cross-validation chose the separated models, & that model also did well on the test set, so it gave a better way to choose between the two approaches.

Problem 4)

4.3) I agree most with the features are weakly predictive of Shaquille O'Neal's freestyle free throws. The logistic regression model did a better than the baseline, so there is some predictive signal in the features. But the improvement was small, so the features don't seem strongly predictive.

4.4)

i) We can't use the final score as feature because the final score is only known after the game ends, but we are trying to predict the free throw at the moment it is taken. So using the final score would be using future information that would not actually be available when making the prediction.

ii) Ideally, we should hold out whole games, as games are related to each other. If some shots from one game are in training & others from that same game are in test, the test set is not really fully new, so the model may look better than it actually is.